PROBLEM STATEMENT:

Write a menu driven program in C to implement the basic arithmetic operations such as Addition, Subtraction, Multiplication, and Division.

OBJECTIVES:

• To understand the concept of menu driven programming in C.

• To apply conditional statements and switch–case control structures.

• To perform basic arithmetic operations using user-defined input.

• To enhance logical thinking and problem-solving skills.

THEORY:

if–else ladder statements syntax and sample program

switch–case or if–else ladder statement syntax and sample progrM

Basic arithmetic operations include Addition, Subtraction, Multiplication, and Division. Division by zero should be avoided.

ALGORITHM:

Type Program

PLATFORM: Linux

CODE:

```
#include <stdio.h>

int main()

{
```

```
    int choice;
    float a, b;
    printf("1. Addition\n2. Subtraction\n3. Multiplication\n4. Division\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    printf("Enter two numbers: ");
    scanf("%f %f", &a, &b);
    switch (choice)
    {
        case 1:
            printf("Addition = %.2f\n", a + b);
            break;
        case 2:
            printf("Subtraction = %.2f\n", a - b);
            break;
        case 3:
            printf("Multiplication = %.2f\n", a * b);
            break;
        case 4:
            if (b != 0)
                printf("Division = %.2f\n", a / b);
            else
                printf("Division by zero is not allowed\n");
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
}
```

Choice of operation and two numbers

```c
#include <stdio.h>

// menu driven arithmetic
int main() {
    int choice;
    float num1, num2;

    printf("Menu:\n");
    printf("1. Addition\n");
    printf("2. Subtraction\n");
    printf("3. Multiplication\n");
    printf("4. Division\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter two numbers: ");
    scanf("%f %f", &num1, &num2);

    switch (choice) {
        case 1:
            printf("Addition = %.2f\n", num1 + num2);
            break;
        case 2:
            printf("Subtraction = %.2f\n", num1 - num2);
            break;
        case 3:
            printf("Multiplication = %.2f\n", num1 * num2);
            break;
        case 4:
            if (num2 != 0)
                printf("Division = %.2f\n", num1 / num2);
            else
                printf("Error: Division by zero\n");
            break;
        default:
            printf("Invalid choice\n");
    }

    // switch case operations
    return 0;
}
```

OUTPUT:

1. Addition

2. Subtraction

3. Multiplication

4. Division

Enter your choice: 1

Enter two numbers: 1 2

Addition = 3.00

Displays the result of the selected arithmetic operation

SAMPLE INPUT:

1

1 2

SAMPLE OUTPUT:

Addition = 3

CONCLUSION:

Thus, the menu driven program in C successfully performs basic arithmetic operations using switch–case statements.

FAQs:

1. What is a menu driven program?

2. Why is switch–case preferred over if–else ladder in menu driven programs?

3. What is the purpose of the break statement in switch–case?

4. What is the purpose of the break statement in switch–case?

5. What are the limitations of switch–case compared to if–else?